

SIDDHARTH PARMAR

Technical Lead – Embedded Systems, Firmware, Enterprise and Software Delivery

Ahmedabad, Gujarat, India | +91 90995 92271 | siddharth.parmar-1993@outlook.com |
linkedin.com/in/siddharth-parmar-986a31a8

PROFESSIONAL PROFILE

Technical Lead with 11 years of experience (since 2015) spanning **embedded systems, firmware development, and enterprise software** delivery. Turns complex, cross-functional engineering programs into predictable, on-time delivery for teams of **10+ engineers and platforms serving 5,000+** end users. Owns execution end-to-end requirement discussions, solution design, sprint execution, stakeholder alignment, and release. At **ATRI Systems**, provides **technical leadership across embedded engineering programs** (firmware, electronics, mechanical, and QA), owning risk, dependencies, and execution discipline from planning through release. Previously spent 5+ years at **Arastu Systems (consulting at Stridely Solutions)** as **Technical Lead** on an **enterprise CPQ platform used by 5,000+** users across the USA and Canada, leading a 10-person engineering team and coordinating with **80+ cross-functional stakeholders through a hybrid Agile-Waterfall model**. Earlier career-built mission-critical **embedded systems for ISRO (SAC & ISTRAC), Nelco, and BSNL antenna control systems, real-time firmware, and hardware-integrated solutions.**

- **Technical leadership:** full SDLC ownership, architecture, code/design review, execution, and release
- **Technical grounding:** C++, Embedded C, C#, .NET, Azure, SQL, RTOS/bare-metal, embedded Linux, and firmware and hardware integration
- **Stakeholder leadership:** large-scale, cross-functional coordination across 80+ stakeholders and 10-person engineering teams
- **Execution discipline:** risk management, delivery governance, and building accountable engineering teams

CORE SKILLS

Technical Leadership: Firmware Architecture, Embedded Systems Design, Code Review, Design Review, Full SDLC Ownership, Legacy Modernization, Cloud Migration

Embedded & Firmware: C/C++, Embedded C, C#/.NET, RTOS (bare-metal & FreeRTOS-class), Embedded Linux, Bootloader & Device Drivers, Real-Time Systems, Multithreading, MISRA-C Awareness, JTAG/Hardware Debugging

Protocols & Interfaces: SPI, I2C, UART, RS232/RS485, SNMP, TCP/IP, UDP, Serial Communication, Hardware/Software Integration

Software & Cloud: Azure, SQL, MFC, VC++, Oracle, Unit Testing, Static Code Analysis, Basic CI/CD Fundamentals

Delivery & Project Management: Agile/Scrum, Hybrid Agile-Waterfall, Sprint & Release Management, Risk & Dependency Management, Execution Governance, Project Planning

Leadership & Stakeholder Management: Team Leadership, Cross-Functional Coordination, Stakeholder Management, Customer Communication, Scope & Priority Alignment

Tools & Methods: Jira, Git, Figma, Draw.io, Technical Documentation, Design Reviews

PROFESSIONAL EXPERIENCE

Technical Lead | ATRI Solutions | Oct 2025 – Present | Gandhinagar

- **Lead end-to-end delivery of embedded engineering programs, driving execution across firmware, electronics, mechanical, QA, and procurement teams;** own project/sprint planning, milestone tracking, risk and dependency management, resource coordination, and stakeholder communication.
- Drive Agile execution through weekly and monthly planning cycles, sprint reviews, and milestone tracking with internal and external stakeholders, keeping delivery on schedule and on scope.
- Act as the technical and delivery bridge between engineering teams and leadership, aligning scope, priorities, execution strategy, and release readiness.

- Proactively identify and resolve cross-functional risks, technical dependencies, and resource constraints before they impact timelines or quality.
- **Provide hands-on technical leadership for embedded firmware architecture, hardware-software integration, and real-time systems, including SPI, I2C, UART, and RS232/RS485 interfaces.**
- Strengthen execution discipline, ownership, and delivery governance across teams, improving predictability and visibility into program health.

Selected Project: Induction-Based Industrial Charging & Heating Platform

- Developed and maintained an **industrial induction-based charging and heating platform that wirelessly transfers energy to a compatible plate/surface and maintains a controlled operating temperature.**
- Owned embedded firmware development and maintenance, control logic, sensor interfacing, system monitoring, fault handling, and performance optimization, including debugging and validation for stable operation.
- Contributed to firmware and system-level integration across induction charging, temperature monitoring, control, and user-interaction components, improving reliability and thermal consistency.

Selected Project: Smart Torque Tool Platform

- Developed and supported a connected smart torque tool combining **embedded electronics, motor control, sensing, an LCD-based UI, and wireless connectivity for real-time monitoring and intelligent tool operation.**
- Worked with BLDC motor control architecture; firmware coordinated motor control, sensor data acquisition, tool logic, and user interactions. Integrated barcode scanning to associate operations with work instructions.
- Implemented **MQTT-based** connectivity for real-time data exchange and remote monitoring, and contributed to over-the-air (OTA) firmware update capability for field maintenance without physical tool access.

Technical Lead | Arastu Systems Pvt. Ltd. (Consultant at Stridely Solutions) | Aug 2024 – Oct 2025 | Ahmedabad

- **Led project planning, technical execution, and end-to-end delivery of enterprise software solutions, defining scope, objectives, milestones, dependencies, and delivery priorities.**
- Managed a team of **10 software developers and coordinated with 80+ cross-functional** stakeholders across engineering, QA, business analysis, project management, delivery, and customer teams.
- Applied a hybrid Agile-Waterfall delivery model, combining structured planning with iterative sprint execution, continuous feedback, and milestone-based delivery.
- Spearheaded modernization of a legacy structural-building quoting and pricing application, transitioning a long-established desktop solution to a modern web-based architecture serving 5,000+ users across the USA and Canada.
- Led full-stack development and technical decision-making across **C++, VC++, MFC, C#, MS SQL, Oracle, and Azure, driving cloud-enabled modernization initiatives.**
- Drove technology evaluation, performance optimization, and scalability improvements while preserving business-critical functionality throughout modernization; collaborated directly with customers, delivery heads, business analysts, and QA to resolve dependencies and manage risk.
- Provided technical leadership across the full SDLC, requirement analysis, architecture discussions, development planning, code-level guidance, integration, testing, release planning, and production delivery.

Selected Project: CPQ Platform Modernization

- **Modernized a CPQ (Configure, Price, Quote) desktop application built on MFC/C++ into an Azure cloud-based web application, transforming a long-standing desktop solution into a scalable, accessible cloud platform while preserving critical business logic and enterprise integrations.**
- Worked primarily on the Quoting Engine, a core component built on VC++ and .NET Framework, incorporating legacy Fortran libraries and integrating with Oracle and MS-SQL databases, requiring careful handling of legacy code and business logic during modernization.
- **Contributed to integrating the desktop application with Azure cloud infrastructure using Virtual Machine Scale Sets (VMSS) and Docker containerization, supporting a scalable, reliable cloud-based architecture.**
- Worked across development, legacy code integration, database connectivity, cloud deployment, troubleshooting, and system integration; the modernized application supports 5,000+ concurrent users across the USA and Canada.

Senior Software Engineer | Arastu Systems Pvt. Ltd. | Aug 2021 – Jul 2024 | Ahmedabad, India

- Specialized in modernizing legacy systems, full-stack development, and performance optimization across C++/VC++, .NET Framework, MFC, and database management, improving efficiency and scalability.

- Managed and mentored the C++ development team, modernizing legacy modules toward web-based applications; engaged directly with clients to track progress and gather requirements.
- Built robust console and dialog-based applications using C++, .NET, and MFC for business-critical use cases; designed and managed Oracle, SQL, and MS Access databases, optimizing structures and queries for performance.
- Implemented multi-threaded and multi-process architectures to improve system efficiency and responsiveness, and developed automation tools/utilities that eliminated repetitive manual tasks.

Software Engineer | Arastu Systems Pvt. Ltd. | Jul 2020 – Jul 2021 | Ahmedabad, India

- Maintained and enhanced a 25-year-old enterprise application using MFC, VC++, and C# (CLR), ensuring reliability, performance, and continuity of business-critical functionality.
- Analyzed, maintained, and troubleshoot legacy applications while implementing improvements that enhanced maintainability and system performance.
- Contributed to application modernization initiatives by introducing newer technologies, development practices, and architectural approaches, and supported migration of desktop applications toward modern web-based solutions.

Project Manager – Sales & Delivery | TechMetix | Nov 2019 – Aug 2020 | Ahmedabad, India

- Managed end-to-end delivery across software, hardware, and IoT-enabled solutions — defining scope and deliverables, preparing execution plans, coordinating resources, and monitoring projects from discovery through final delivery.
- Supported sales and business development through requirement analysis, technical proposals, client presentations, and pre-sales coordination.
- Managed RFID-based school attendance solutions integrating GSM, Ethernet, and Wi-Fi connectivity with centralized web-based monitoring; oversaw hardware development including schematic design, BOM preparation, and prototype validation.

Selected Project: Smart School Attendance System

Developed and managed an RFID-based smart school attendance system that automated student identification and attendance capture, transmitting real-time records to a centralized web portal.

Integrated RFID readers with GSM, Ethernet, and Wi-Fi connectivity options; worked across embedded devices, communication modules, device configuration, web-portal integration, and system validation.

Coordinated requirements, hardware/software development, testing, and deployment to replace manual attendance processes, improve data accuracy, reduce administrative effort, and provide remote access to attendance information.

Head – Controls Design | Sumeru Microwave Communications Pvt. Ltd. | May 2018 – Oct 2019 | Ahmedabad, India

- Led design and development of 3-axis motor-based Antenna Control Units (ACU) supporting induction, stepper, and servo motors, with local and remote control capabilities.
- Directed embedded software development including custom communication protocols, AVR ATmega microcontroller programming, and GUI-based control applications for Windows IoT Core, UWP, and desktop platforms.
- Owned hardware development — schematic design, 2-layer PCB layout, BOM preparation, enclosure design, and prototype validation; implemented RS232, RS485, Ethernet (TCP/IP), and SNMP for reliable system communication.

Selected Project: ACU – Antenna Control Unit / System

Designed and developed a microcontroller-based 3-axis Antenna Control Unit (ACU) for precision antenna positioning using servo motors with absolute optical encoders in a closed-loop control architecture.

Developed embedded control logic for real-time motor control, encoder feedback processing, system monitoring, and control operations; integrated a 7-inch touchscreen local UI for configuration, monitoring, and manual control.

Implemented local and remote control with custom communication protocols and SNMP-based monitoring/control, integrating embedded firmware, hardware interfaces, UI, testing, validation, troubleshooting, and technical documentation.

Selected Project: Motor-less Antenna Control & Monitoring System

Developed a motor-less antenna control and monitoring system for antenna positioning, tracking, and visualization using an inbuilt GPS module and angle-measurement sensors.

Processed GPS and orientation data in the embedded controller to provide real-time antenna status and tracking information; developed a UWP graphical interface with pictorial satellite-tracking visualization.

Integrated the embedded controller, GPS, angle sensors, communication interfaces, and Windows-based UWP application into an end-to-end monitoring solution focused on simplicity and usability.

Engineer – Control Systems | Sumeru Microwave Communications Pvt. Ltd. | Aug 2015 – Apr 2018 | Ahmedabad, India

- Developed embedded firmware using Embedded C and AVR/ATmega microcontrollers for 3-axis antenna control systems, including control algorithms, system logic, and firmware integration.
- Designed custom communication protocols and electronic schematics/2-layer PCB layouts; implemented RS232, RS485, Ethernet/TCP-IP, and SNMP for system control and monitoring.
- Developed Raspberry Pi 3B+ based touchscreen HMI solutions using Windows IoT Core and UWP for embedded antenna control and monitoring applications.

Selected Project Experience: Embedded Antenna Control & HMI Solutions

Developed 3-axis motor-based antenna control systems supporting induction, stepper, and servo motor technologies with local and remote control capabilities.

Developed Embedded C firmware and AVR/ATmega microcontroller solutions including control algorithms, system logic, custom communication protocols, and firmware integration.

Integrated embedded controllers with motors, sensors, GPS, and Raspberry Pi 3B+ touchscreen HMI solutions using Windows IoT Core and UWP; implemented RS232, RS485, Ethernet/TCP-IP, and SNMP for control and monitoring.

EDUCATION & CERTIFICATIONS

B.E. – Electronics & Communication | Indus Institute of Technology & Engineering – GTU, Ahmedabad | 2015

- AI for Project Management: Boosting Collaboration with Generative AI — LinkedIn Learning (PMI REP), Aug 2026
- Microsoft Azure Fundamentals (AZ-900) — LinkedIn Learning, Apr 2025
- C++ Advanced Concepts & Best Practices — LinkedIn Learning, Apr 2025
- Transitioning from Tech to Manager — LinkedIn Learning, Jun 2025

LANGUAGES

English, Hindi, Gujarati